

TITLE PAGE

Problem Statement ID- SIH26007

Problem Statement Title- Safe and Efficient Operation of Mine Vehicles in Fog and Low-Visibility Conditions in Open Cast Iron Ore Mines

Theme- Smart Automation

PS Category- Hardware

Team ID-

Team Name (Registered on portal) - Wired Differently



IDEA TITLE

SMARTFOG — Turning Low Visibility into Actionable Situational Awareness

When fog takes away visibility, SmartFog reconstructs the information the driver needs to know.

How SmartFog Works?

SmartFog transforms fragmented sensor data into real-time, actionable safety awareness for mine vehicles in low visibility.

01 SENSE

SmartFog gathers real-time information from the vehicle's camera, distance sensors, vehicle telemetry, and GPS/DGPS with digital mine-map data. These inputs provide the raw information required to understand the vehicle's surroundings, position, movement, and road context even under low-visibility conditions.

02 PERCEIVE

The system processes these inputs to identify nearby vehicles, obstacles, visibility conditions, distance, and relative movement. AI-based perception continuously interprets the environment and converts raw sensor readings into meaningful information about what is ahead and how the surrounding situation is changing.

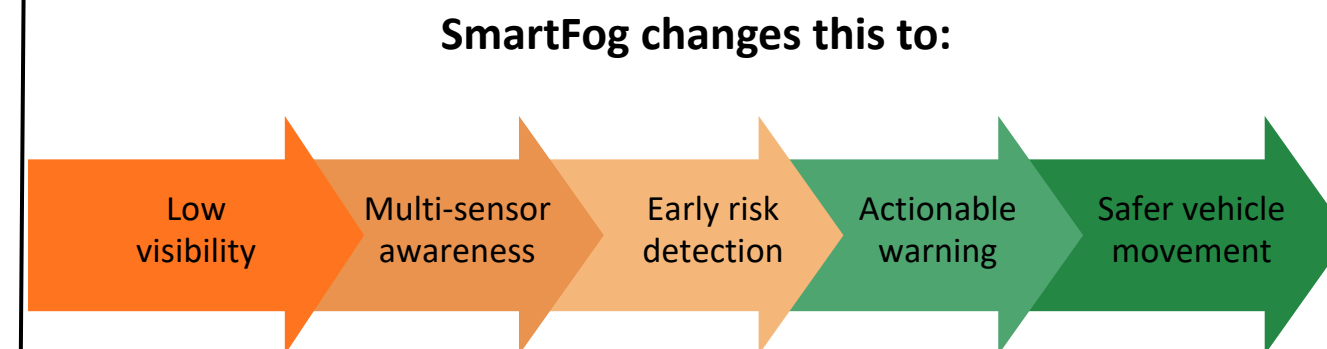
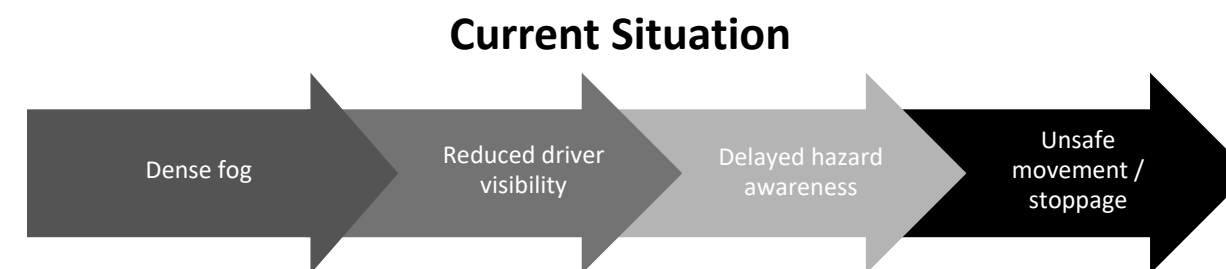
03 ASSESS

SmartFog combines camera, sensor, vehicle, positioning, and map information to build a more reliable picture of the situation. The risk engine evaluates distance, speed, visibility, object type, relative movement, and road geometry to estimate collision risk and Time-to-Collision (TTC) in real time.

04 ACT

The system converts the assessed risk into a simple, actionable safety recommendation for the driver and control room. Depending on the situation, SmartFog can indicate CONTINUE, SLOW DOWN, or STOP, while sharing critical risk information across the fleet for coordinated safety intervention.

How it addresses the problem?



- Gives the driver hazard awareness even when direct visibility is poor
- Detects vehicles, obstacles, and unsafe proximity in real time
- Estimates distance and relative movement of detected objects
- Recommends speed reduction or stopping based on assessed risk
- Provides advance awareness of upcoming turns and road geometry
- Alerts drivers about oncoming vehicles and potential collision risks
- Converts multiple sensor inputs into simple actionable safety decisions

Uniqueness and Innovation

SmartFog does not claim to “see through fog.” It reconstructs situational awareness using multiple information sources.



Multi-Sensor Fusion

Combines camera, distance, vehicle, and positioning data for reliable hazard awareness



Risk-Based Decision Engine

Analyzes real-time hazards and recommends appropriate actions before situations become critical.



Map-Assisted Road Awareness

Uses digital mine maps and vehicle position to anticipate turns and road geometry



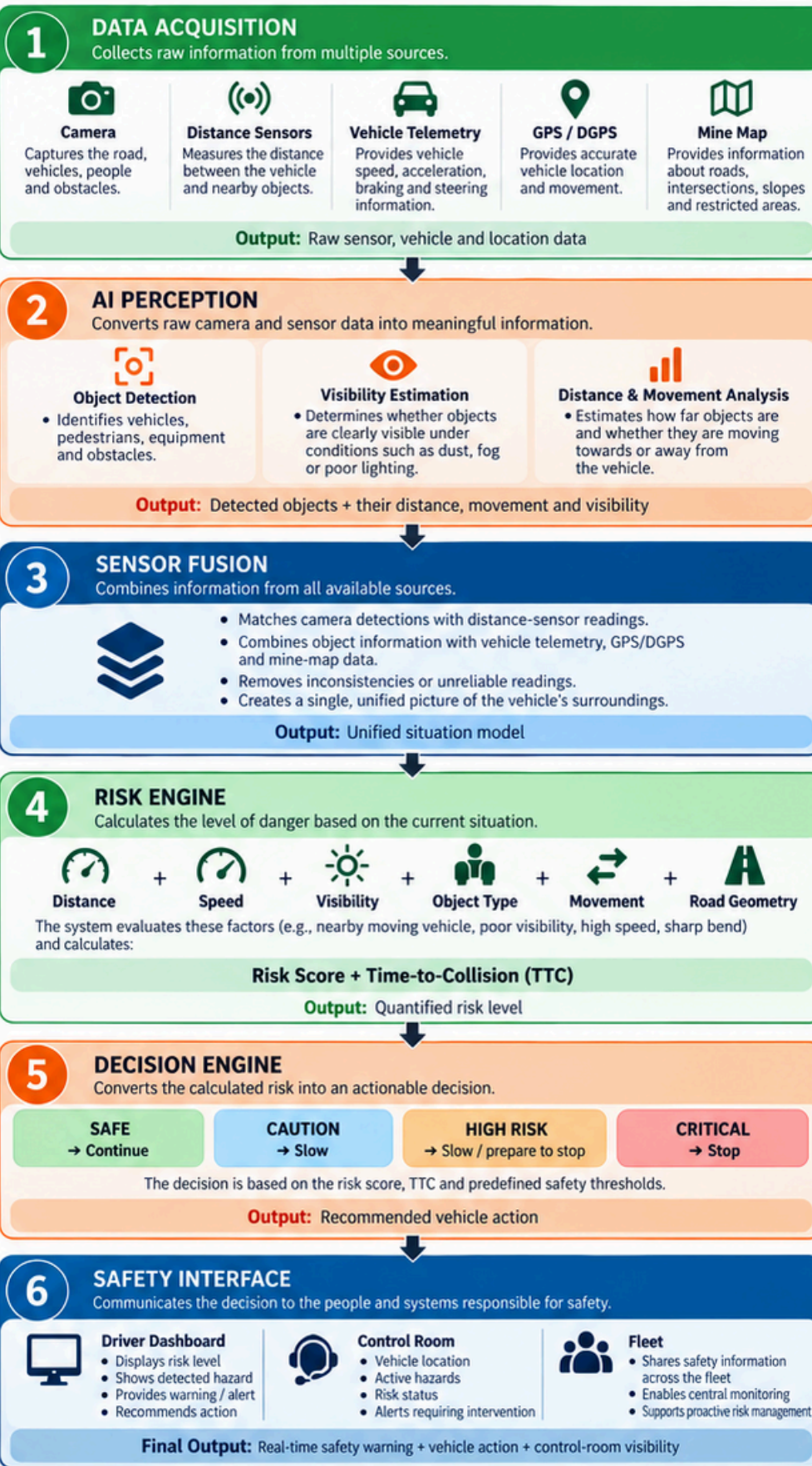
Fleet-Wide Safety Intelligence

Connects multiple dumpers and control rooms for coordinated, real-time safety monitoring.

TECHNICAL APPROACH

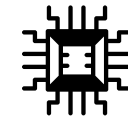
System Architecture

Technology Stack



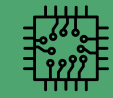
Software & AI

Python • OpenCV • YOLO/CV Model • Flask/FastAPI • HTML/CSS/JavaScript
AI perception, visibility estimation, risk assessment and dashboard processing.



Embedded & Communication

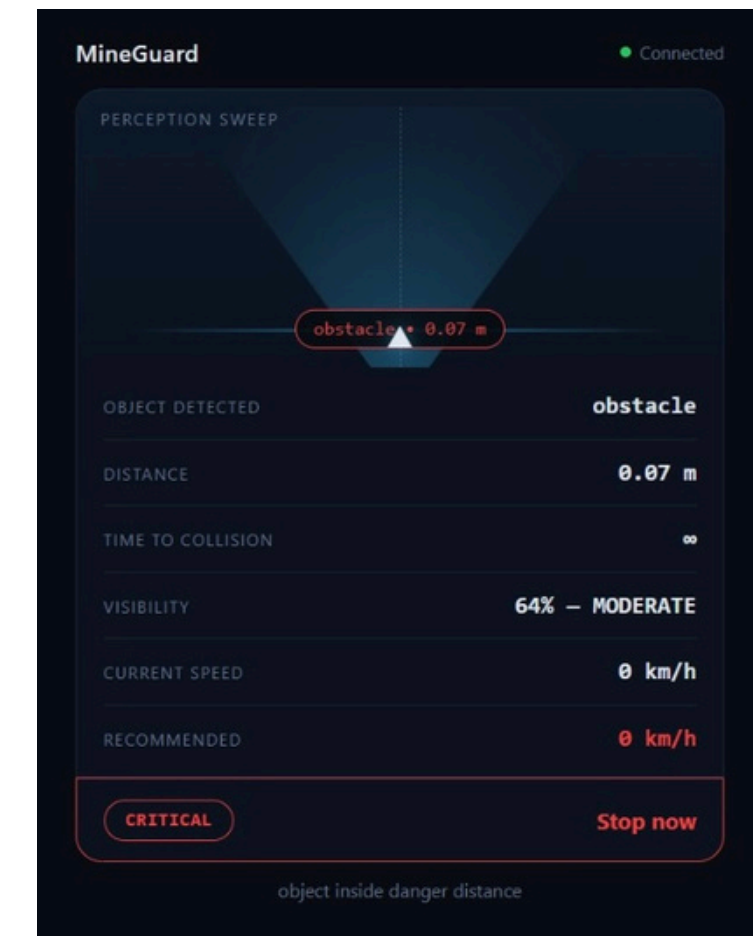
C/C++ • Arduino IDE • ESP32 • ESP32-CAM • Wi-Fi
Vehicle control, sensor acquisition, camera streaming and prototype communication.



Hardware & Sensors

ESP32 DevKit • ESP32-CAM • HC-SR04 • L298N • 4WD Chassis • DC Motors
Prototype platform for distance sensing, vehicle movement and real-time camera data.

Wireframes



Prototype: ESP32 + ESP32-CAM + HC-SR04 → Laptop AI Processing → Driver/Control Room Dashboard

FEASIBILITY AND VIABILITY

Feasibility of the Solution

Technically Feasible

- Prototype uses readily available embedded hardware and sensors.
- AI perception and risk processing can run on edge/laptop systems.
- Modular architecture allows additional sensors to be integrated later.
- GPS/DGPS and digital mine maps can extend awareness beyond camera visibility.
- Existing prototype demonstrates vehicle control, sensing, camera streaming and risk-based decisions.

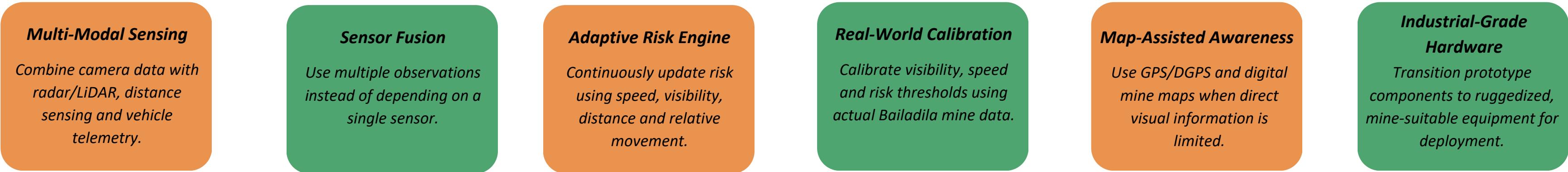
Deployment Feasibility

- Can begin with a single-vehicle pilot before fleet deployment.
- Prototype architecture can evolve toward industrial-grade hardware.
- Existing mine fleet infrastructure can be integrated progressively.

Potential Challenges and Risks

Challenge / Risk	Why it matters
Extreme fog / poor camera visibility	Camera perception can become unreliable
Sensor limitations	Individual sensors may produce incomplete data
False alarms	Excessive warnings can reduce driver trust
GPS/GNSS limitations	Position accuracy may degrade in difficult environments
Prototype vs. mine conditions	Lab thresholds may not represent real conditions
Communication reliability	Fleet coordination requires dependable connectivity
Industrial ruggedness	Mine environments demand durable hardware

Strategies for Overcoming Challenges



IMPACT AND BENEFITS

SAFETY

- Earlier awareness of vehicles, obstacles and collision risks
- Real-time speed, stop recommendations based on risk
- Improved driver awareness during low visibility
- Reduced unsafe proximity and potential collision events

PRODUCTIVITY

- Supports safer vehicle movement during low-visibility conditions
- Helps reduce avoidable fog-related stoppages
- Enables more predictable haul cycles
- Improves utilization of mine vehicles and haul roads

FLEET AND OPERATIONS

- Supports safer vehicle movement during low-visibility conditions
- Helps reduce avoidable fog-related stoppages
- Enables more predictable haul cycles
- Improves utilization of mine vehicles and haul roads

ECONOMIC & ENVIRONMENTAL

- Reduces productivity losses caused by unnecessary stoppages
- Helps reduce vehicle idling and inefficient movement
- Supports better fleet utilization and operational efficiency
- Creates a foundation for safer, smarter and more sustainable mining

RESEARCH AND REFERENCES

NMDC Limited — Problem Statement, Smart India Hackathon 2026 (SIH26007): “Safe and Efficient Operation of Mine Vehicles in Fog and Low-Visibility Conditions in Open Cast Iron Ore Mines.”

Ultralytics YOLO Documentation — <https://docs.ultralytics.com>

OpenCV Documentation — <https://docs.opencv.org>

Espressif Systems — ESP32 Technical Reference Manual, <https://www.espressif.com>

HC-SR04 Ultrasonic Distance Sensor — Datasheet

L298N Dual H-Bridge Motor Driver — Datasheet

Ministry of Steel, Government of India — National Steel Policy (referenced for NMDC production targets)